



DATABASE SYSTEM LABS

- BASIC SQL III

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OBJECTIVES

- Basic Queries
- Natural join
- Rename operation
- String operation
- Ordering the display of tuples
- Set operation
- Aggregate Functions
- Nested Subqueries

Basic Queries

BASIC QUERIES

- `select * from course;`
- `select name`
 `from instructor`
 `where dept_name = 'Comp. Sci.';`
- `select name, course_id`
 `from instructor , teaches`
 `where instructor.ID = teaches.ID`

Natural join

NATURAL JOIN

- `select * from teaches natural join course;`
- `select * from teaches join course on teaches.course_id = course.course_id;`

Rename operation

NATURAL JOIN

- `select avg(budget) as avg_budget from department;`
- `select distinct T.name
from instructor as T, instructor as S
where T.salary > S.salary and S.dept_name = 'Comp. Sci.'`

String operation

STRING OPERATION

- select name from instructor where name like '%on%'
- select * from department where dept_name like '_in%'

Ordering the display of tuples

ORDERING THE DISPLAY OF TUPLES

- select distinct name
from instructor
order by name
- select *
from instructor
order by salary desc;

Set operation

SET OPERATION

- UNION

- Find courses that ran in Fall 2009 or in Spring 2010

(select course_id from section where semester = 'Fall'
and year = 2009)

union

(select course_id from section where semester = 'Spring'
and year = 2010)

SET OPERATION

- INTERSECT

- Find courses that ran in Fall 2009 and in Spring 2010
(select course_id from section where semester = 'Fall'
and year = 2009)
intersect
(select course_id from section where semester = 'Spring'
and year = 2010)

SET OPERATION

- EXCEPT

- Find courses that ran in Fall 2009 but not in Spring 2010

(select course_id from section where semester = 'Fall'
and year = 2009)

except

(select course_id from section where semester = 'Spring'
and year = 2010)

Aggregate Functions

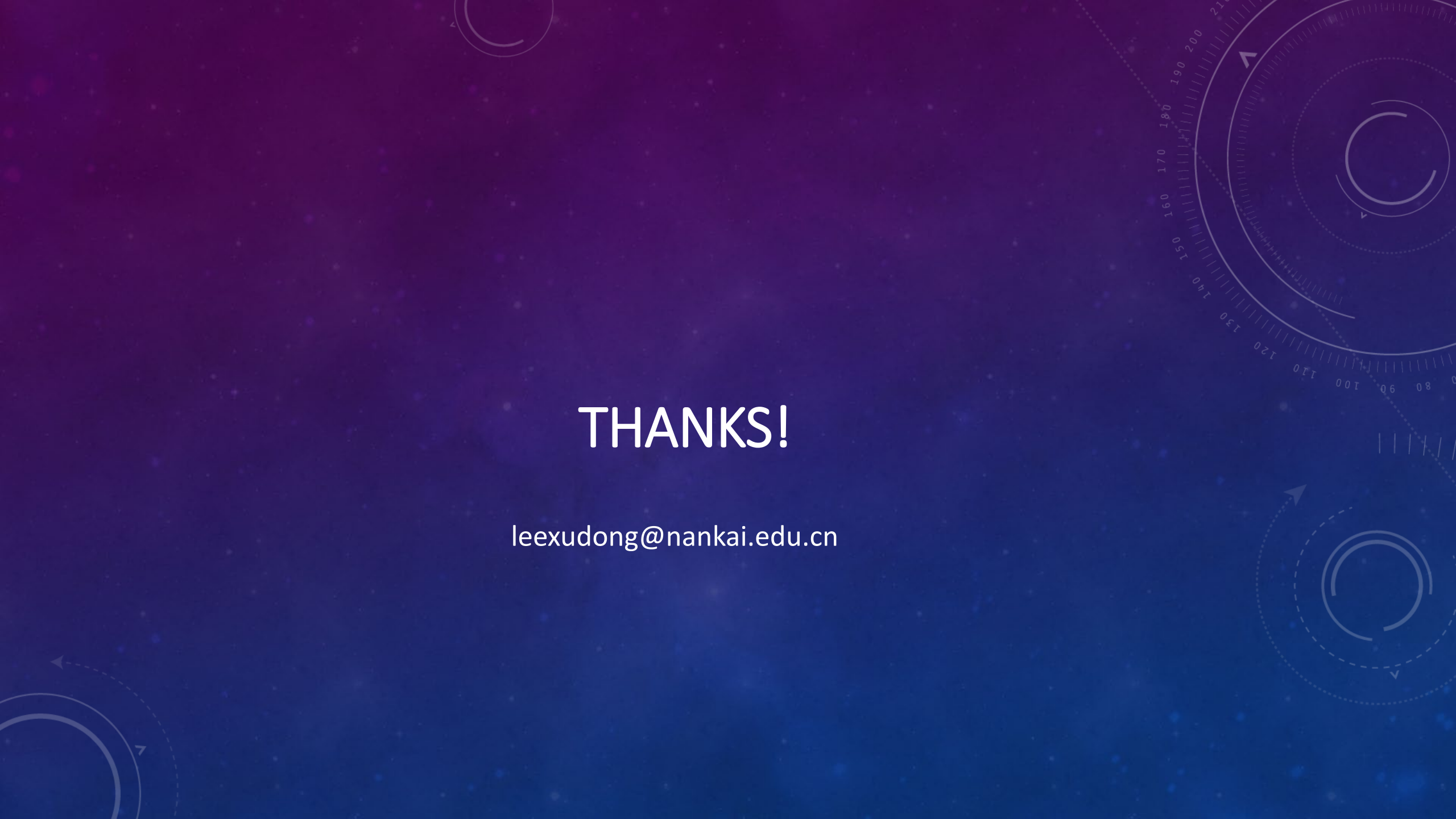
AGGREGATE FUNCTIONS

- `select dept_name, avg (salary)`
`from instructor`
`group by dept_name`
`having avg (salary) > 42000;`

Nested Subqueries

NESTED SUBQUERIES

- `select count(distinct ID)`
 `from takes`
 `where (course_id, sec_id, semester, year)`
 `in (select course_id, sec_id, semester, year`
 `from teaches`
 `where teaches.ID= 43779);`



THANKS!

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